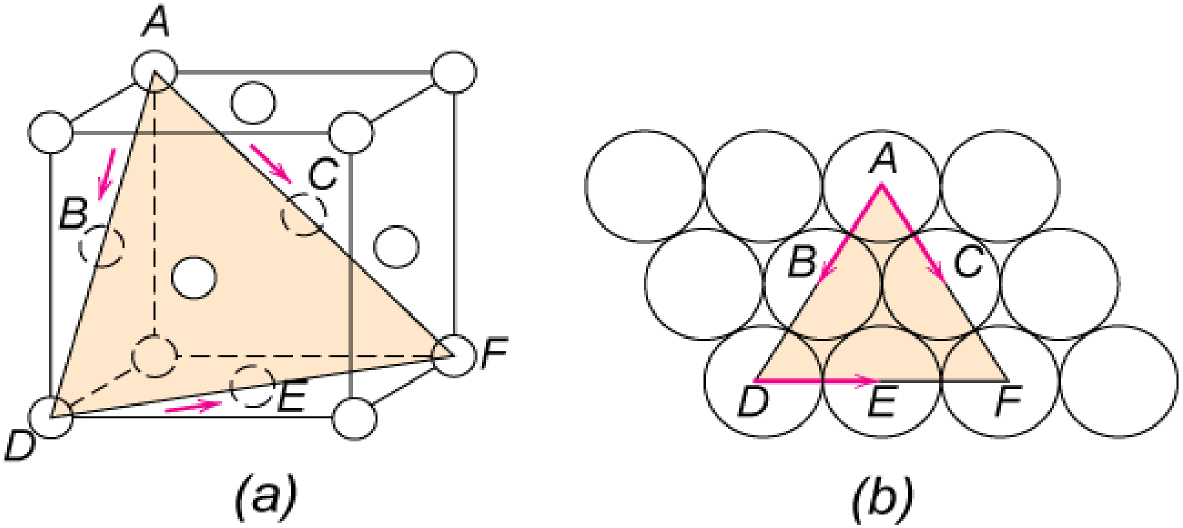
Deformation Mechanisms

Slip System

– Slip plane - plane allowing easiest slippage
  - Wide interplanar spacings - highest planar densities

– Slip direction - direction of movement - Highest linear densities



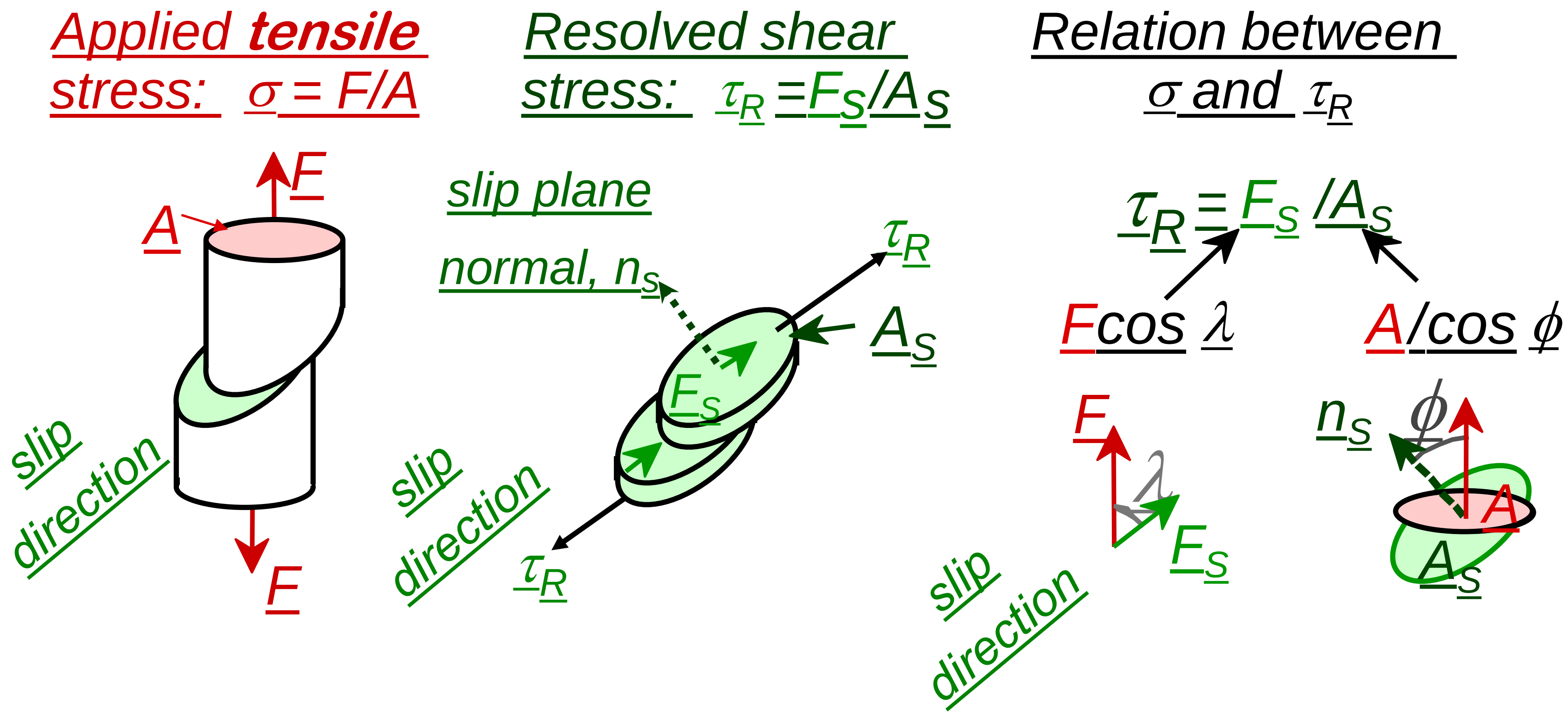
– FCC Slip occurs on {111} planes (close-packed) in <110> directions (close-packed)

=> total of 12 slip systems in FCC

– in BCC & HCP other slip systems occur

Stress and Dislocation Motion

- Crystals slip due to a resolved shear stress, τ_R.
- Applied tension can produce such a stress.



$$\tau_R = \sigma \cos\lambda \cos\phi$$

Critical Resolved Shear Stress

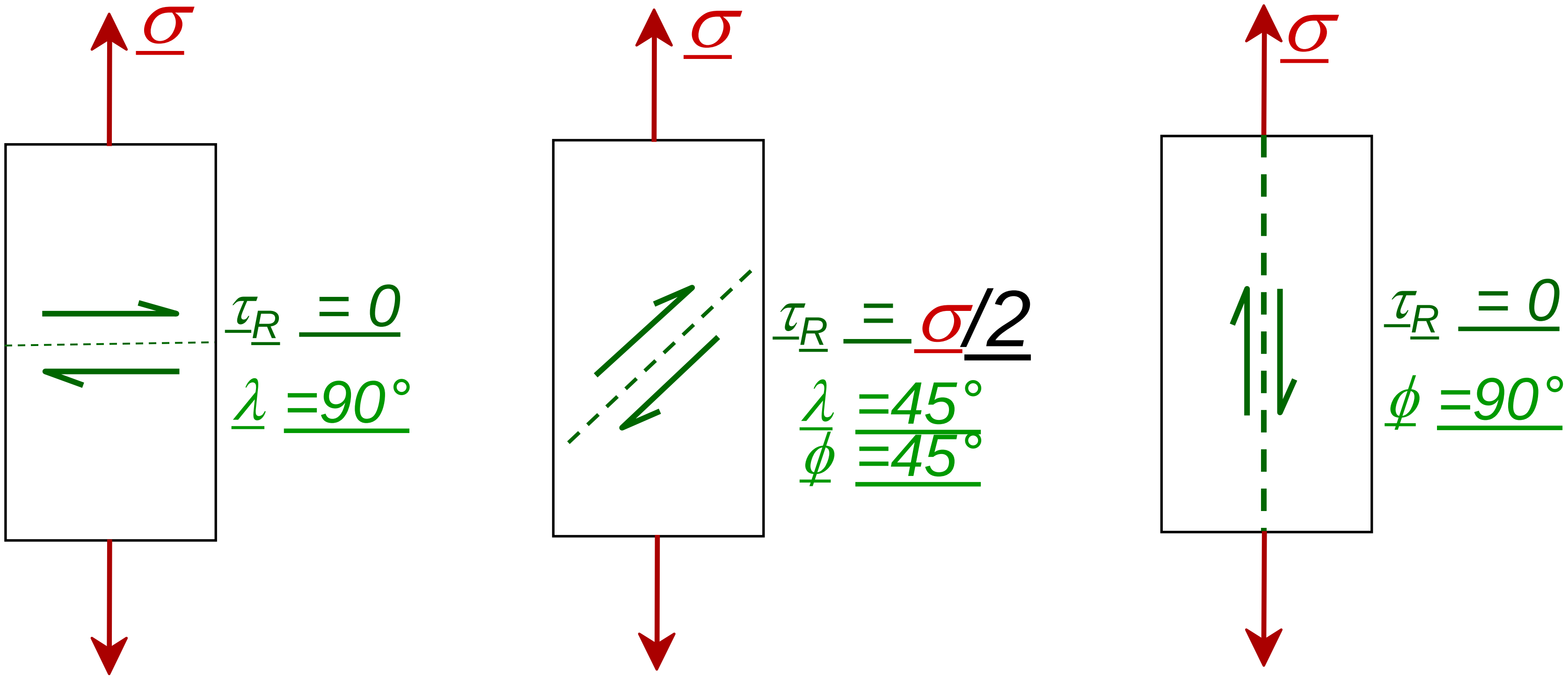
- Condition for dislocation motion: $\tau_R > \tau_{CRSS}$

typically

10^{-4} GPa to 10^{-2} GPa

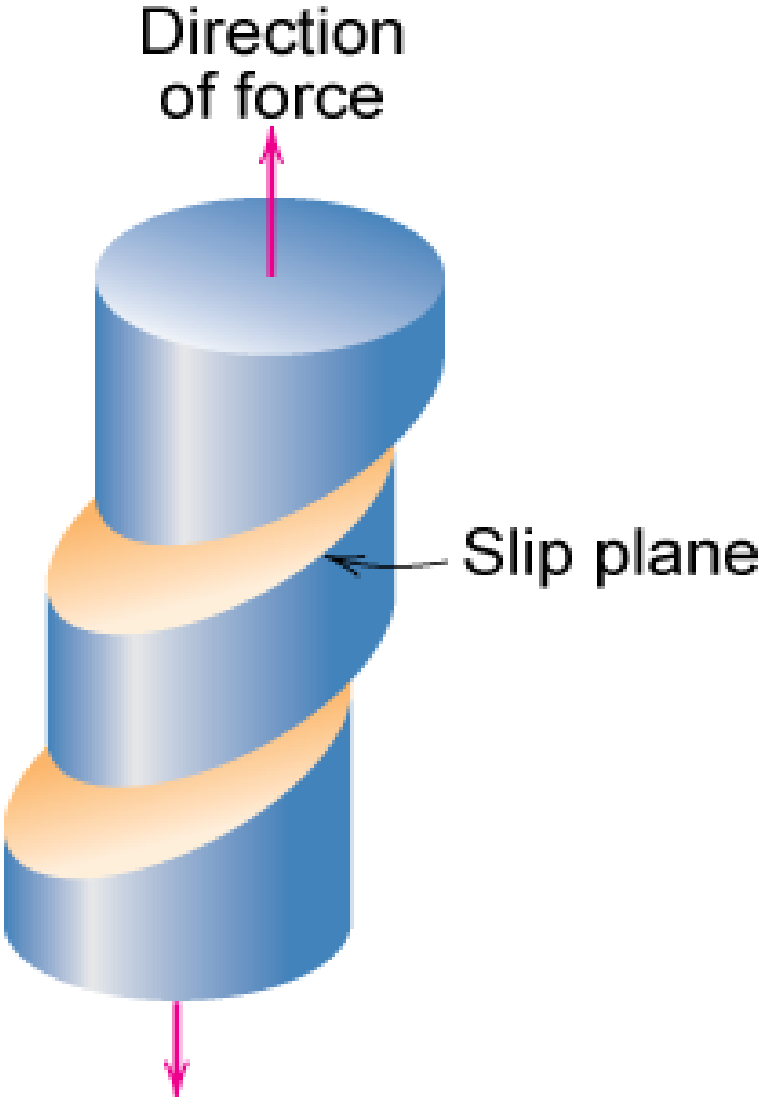
- Crystal orientation can make it easy or hard to move dislocation

$$\tau_R = \sigma \cos\lambda \cos\phi$$

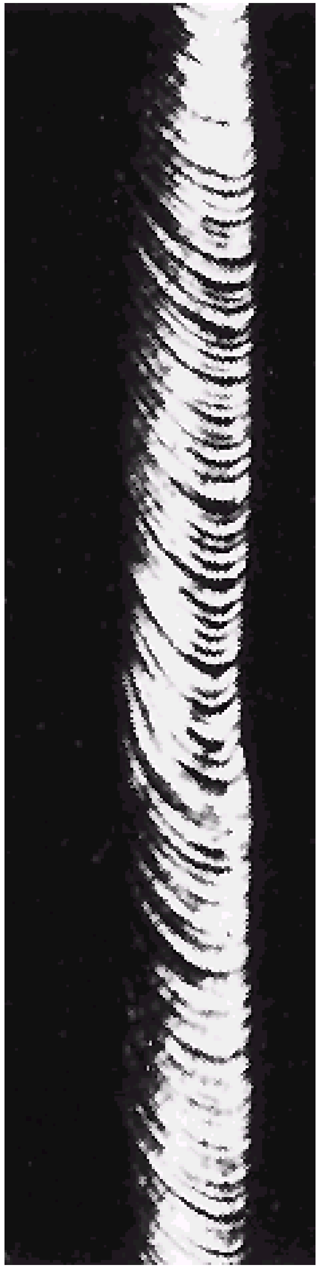


τ maximum at $\lambda = \phi = 45^o$

Single Crystal Slip

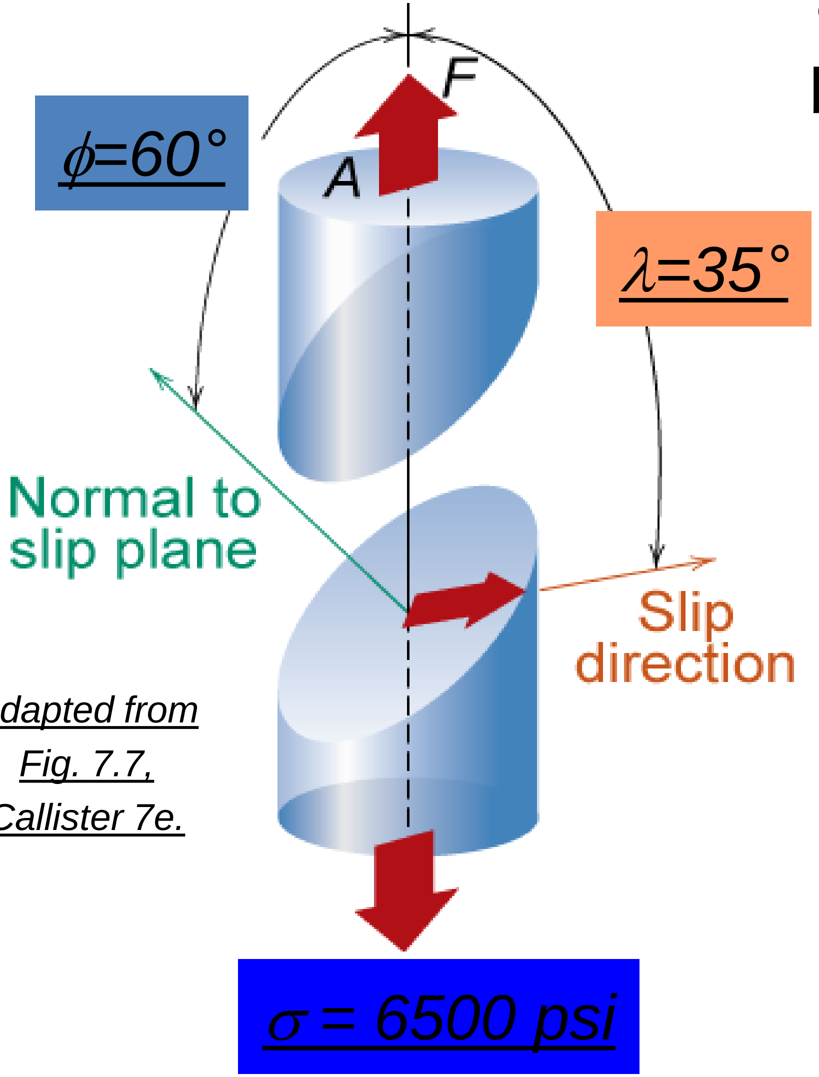


Adapted from Fig. 7.8, Callister 7e.



Adapted from Fig. 7.9, Callister 7e.

Ex: Deformation of single crystal



Adapted from Fig. 7.7, Callister 7e.

a) Will the single crystal yield?

b) If not, what stress is needed?

τ_{crss} = 3000 psi

$$\tau = \sigma \cos\lambda \cos\phi$$

$$\sigma = 6500 \text{ psi}$$

$$\tau = (6500 \text{ psi})(\cos 35^\circ)(\cos 60^\circ)$$
$$= (6500 \text{ psi})(0.41)$$

$$\boxed{\tau = 2662 \text{ psi} \quad < \tau_{crss} = 3000 \text{ psi}}$$

So the applied stress of 6500 psi will not cause the crystal to yield.

Ex: Deformation of single crystal

What stress is necessary (i.e., what is the yield stress, σ_y)?

$$\tau_{crss} = 3000\ \text{psi} = \sigma_y \cos\lambda \cos\phi = \sigma_y (0.41)$$

$$\therefore \sigma_y = \frac{\tau_{crss}}{\cos\lambda \cos\phi} = \frac{3000\ \text{psi}}{0.41} = \underline{\underline{7325\ \text{psi}}}$$

So for deformation to occur the applied stress must be greater than or equal to the yield stress

$$\boxed{\sigma \geq \sigma_y = 7325\ \text{psi}}$$

Slip Motion in Polycrystals

- Stronger - grain boundaries pin deformations
- Slip planes & directions (λ, ϕ) change from one crystal to another.
- τ_R will vary from one crystal to another.
- The crystal with the largest τ_R yields first.
- Other (less favorably oriented) crystals yield later.

Adapted from Fig. 7.10, Callister 7e. (Fig. 7.10 is courtesy of C. Brady, National Bureau of Standards [now the National Institute of Standards and Technology, Gaithersburg, MD].)

Cross-slip

- Consider a screw dislocation moving on one slip plane that encounters an obstacle and is blocked from further movement. This dislocation can shift to a second intersecting slip system, also properly oriented, and continue to move. This is called **cross-slip.**

- In many HCP metals, no cross-slip can occur because the slip planes are parallel (i.e., not intersecting). Therefore, polycrystalline HCP metals tend to be brittle.

- Fortunately, additional slip systems become active when HCP metals are alloyed or heated, thus improving ductility.

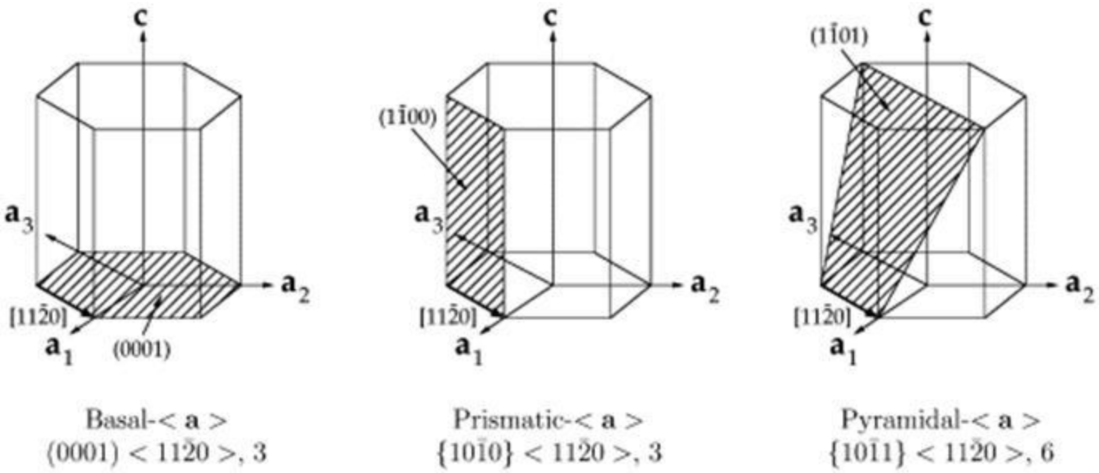


Image Ref: A. Staroselsky, L. Anand, International Journal of Plasticity **2003, 19, 1843**

- Cross-slip is possible in both FCC and BCC metals because a number of intersecting slip systems are present. Consequently, cross-slip helps maintain ductility in these metals.

Influence of crystal structure – number of slip systems

- FCC metals contain four nonparallel close-packed planes of the form {111} and three close-packed directions of the form 110 within each plane, giving a total of **twelve slip systems**. At least one slip system is favorably oriented for slip to occur at low applied stresses, ***permitting FCC metals to have high ductilities***.

- For BCC metals have as many as **48 slip systems** that are nearly close-packed. Several slip systems are always properly oriented for slip to occur, ***allowing BCC metals to have ductility***.

- Where as an ideal HCP metals possess only one set of parallel close-packed planes, the (0001) planes, and three close-packed directions, giving ***three slip systems***. Consequently, the probability of the close-packed planes and directions being oriented with and near 45° is very low. ***The HCP crystal may fail in a brittle manner without a significant amount of slip.***

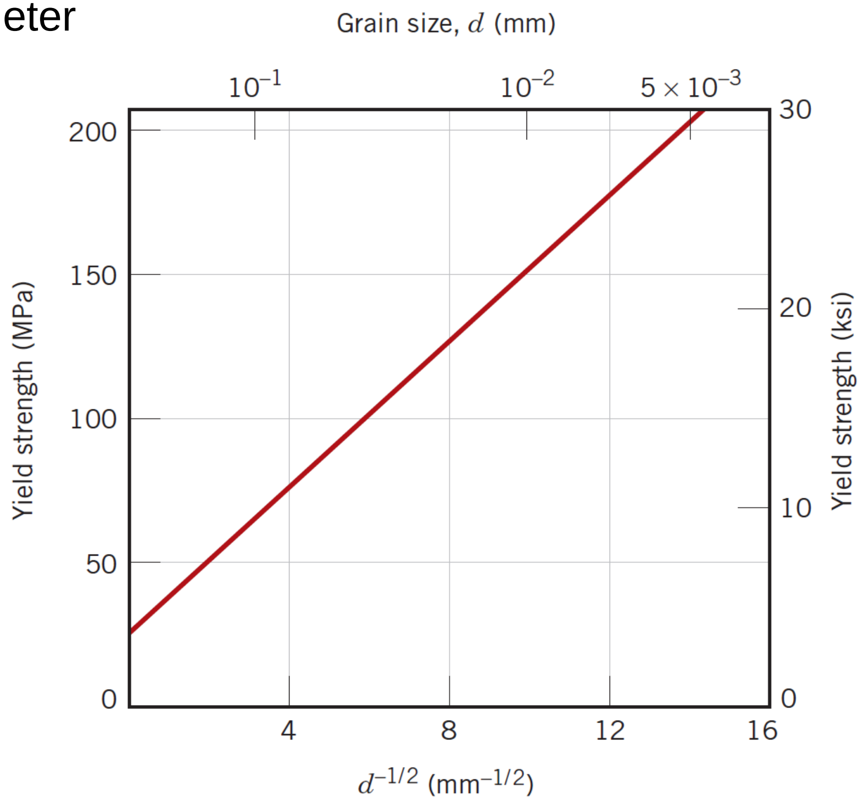
- **Note:** In HCP metals with a low *c/a ratio, or when* HCP metals are properly alloyed, or when the temperature is increased, other slip systems become active, making these metals less brittle than expected.

Strengthening of materials - Hall-Petch equation

- The Hall-Petch equation describes the change in the yield strength of a polycrystalline material with a change in the size of the grains.
- The relationship between yield strength and grain size in a metallic material is given by,

$$\boxed{\sigma_y = \sigma_o + kd^{-1/2}}$$

where σ_y is yield strength; σ_o is the stress required for dislocation movement; K is the material constant, and d is the average grain size diameter



Note: The yield strength increases as the square root of the grain size decreases. Thus, as grains get smaller, the yield strength becomes larger. (The reason for this is that, with smaller grain sizes, the polycrystalline material will have more grain boundaries per unit volume that can act as pinning centers to hold the dislocations in place, **N = 2^{n-1}**)

Hall-Petch equation is a method of strengthening materials by changing their average crystallite (grain) size.

Problem

The yield strength of mild steel with an average grain size of 0.05 mm is 20,000 psi. The yield stress of the same steel with a grain size of 0.007 mm is 40,000 psi. What will be the average grain size of the same steel with a yield stress of 30,000 psi? Assume the Hall-Petch equation is valid and that changes in the observed yield stress are due to changes in dislocation density.

Solution

for a grain size of 0.05 mm the yield stress is 20 x 6.895 MPa = 137.9 MPa, Using the Hall-Petch equation,

$$\boxed{137.9 = \sigma_o + \frac{k}{\sqrt{0.05}}}$$

[1000 psi = 6.895 MPa]

for a grain size of 0.07 mm the yield stress is 40 x 6.895 MPa = 275.8 MPa, Using the Hall-Petch equation,

$$\boxed{275.8 = \sigma_o + \frac{k}{\sqrt{0.007}}}$$

Solving these two equations K = 18.43 MPa·mm$^{1/2}$, and σ_o = 55.5 MPa, and Now, the Hall-Petch equation,

$$\boxed{\sigma_y = 55.5 + 18.43\ d^{-1/2}}$$

If we want a yield stress of 30,000 psi, then

30 x 6.895 = 206.9 MPa, the grain size will be **0.0148 mm**.